

COIT20246 Cyber Security and Networking

Project Specification 1

**Small Business Network Security, OpenWrt Firewall Configuration and Cyber
Security Risk Assessment**

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**Business Scenario: Small IT and Business Consultancy, Brisbane, Queensland,
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4.2.1 Harden the OpenWRT System

Figure 1: Successful SSH Login to the OpenWrt Virtual Machine

Figure 2: Successful Root Password Change on OpenWrt

Figure 3: OpenWrt Password Hashes Stored in /etc/shadow

Disabling unnecessary services reduces the system's attack surface by removing functions that are not required. Fewer active services mean fewer open ports, processes, and potential vulnerabilities for attackers to exploit. This limits possible entry points, reduces security risks, and makes the OpenWrt system easier to monitor and maintain securely.

Hardening Step	Security Risk Addressed
1. Change Default Root Password	Reduces the risk of unauthorised administrative access through default or easily guessed credentials. A strong, unique password makes credential-based attacks more difficult.
2. Examine Password Storage in /etc/shadow	Addresses the risk of password disclosure . Storing passwords as hashes rather than plaintext prevents the original passwords from being directly exposed if the password database is accessed.
3. Set Up SSH Key-Based Authentication	Reduces the risk of password guessing and brute-force attacks . SSH keys provide stronger authentication than password-only access, particularly when a passphrase protects the private key.
4. Disable Unnecessary Services	Reduces the attack surface of OpenWrt. Unnecessary services can provide additional entry points that an attacker could exploit.

4.2.2 Capture and Analyse Network Traffic

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root@OpenWrt:/#
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root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/# tcpdump -i br-mng -s 0 -w /tmp/http_capture.pcap host 192.168.56
.1 and tcp port 80
[ 6929.479623] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144 b
ytes

^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
[ 7069.406559] device br-mng left promiscuous mode
root@OpenWrt:/#

```

Figure 6: HTTP Traffic Capture Using tcpdump on the OpenWrt br-mng Interface

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root@OpenWrt:/#
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root@OpenWrt:/#
root@OpenWrt:/# tcpdump -i br-mng -s 0 -w /tmp/http_capture.pcap host 192.168.56
.1 and tcp port 80
[ 6929.479623] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144 b
ytes

^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
[ 7069.406559] device br-mng left promiscuous mode
root@OpenWrt:/# S_

```

Figure 7: HTTP Traffic Capture Using tcpdump with Host and TCP Port Filtering

9	22.000032	192.168.56.1	192.168.56.255	UDP	82 64846 → 1947 Len=40
10	22.001480	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
11	22.488650	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
12	23.002769	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
13	23.488548	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
14	28.312307	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1
15	29.312964	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1
16	30.313835	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1
17	31.314423	192.168.56.1	239.255.255.250	SSDP	212 M-SEARCH * HTTP/1.1

Figure 8: Captured Network Traffic Displayed in Wireshark (HTTP)

The attacker who taps information from an HTTP stream is able to obtain the server and client IP addresses, the request and response URL, HTTP requests and responses, and the content of the target web pages. Personal information, including names, student IDs or details that could be seen from the webpage, might be given away due to lack of encryption in the HTTP. Any traffic monitoring, and information disclosure, would be possible.

```

ip: can't find device br-mng
root@OpenWrt:~# tcpdump -i br-mng -s 0 -w /tmp/ssh_capture.pcap host 192.168.56.
1 and tcp port 22
[ 7647.756077] device br-mng entered promiscuous mode
tcpdump: listening on br-mng, link-type EN10MB (Ethernet), capture size 262144 b
ytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
[ 7691.350391] device br-mng left promiscuous mode
root@OpenWrt:~#

```

Figure 9: SSH Traffic Capture Attempt Using tcpdump on OpenWrt

56	126.282208	0a:00:27:00:00:04	LLDP_Multicast	LLDP	65 LA/ADMINISTRATOR MA/0a:00:27:00:00:04 3601
57	131.606133	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
58	132.000284	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
59	132.606697	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
60	133.000242	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
61	142.606635	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
62	143.000234	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.168.100.1
63	143.607263	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources
64	144.000388	192.168.56.1	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.168.100.1 for any sources

Figure 10: SSH Network Traffic Packets Displayed in Wireshark During Traffic Analysis

In the HTTP capture it is possible to read webpage content and requests. Encrypted protocols like SSH, on the other hand, keep contents of the sessions secure from being viewed directly. Encryption ensures the confidentiality of the data being sent, minimising the chances of attackers being able to obtain commands, credentials or any other sensitive information through network interception.